

Adib El Dada

Hamilton, Ontario, Canada | (365) 384-0330 | adib.eldada@gmail.com
linkedin.com/in/adib-el-dada-164b29346 | github.com/adibeldada | adibeldada.com

EDUCATION

McMaster University

Sept. 2024 – May 2029

Bachelor of Engineering & Management – Software Engineering & Management

Level III, Co-op

- Program GPA: **3.7/4.0** (10.2/12), **Dean's Honour List** (2025, 2026), **McMaster Award of Excellence**
- Relevant Coursework: Digital Systems & Interfacing (**A+**), Computer Architecture (**A+**), Object-Oriented Programming (**A+**), Intro to Software Development (**A-**), Software Eng. Practice & Experience (**A-**)

TECHNICAL SKILLS

Languages: C, C++, Java, Python, Verilog, TypeScript, JavaScript, Bash, SQL

Hardware & Embedded: FPGA (DE1-SoC), Quartus Prime, Arduino, Raspberry Pi, Digital Logic & FSM Design

Frameworks: React, Next.js, Node.js, Flask, Tailwind CSS, REST APIs, JWT Authentication

Tools: Git, GitHub Actions, Docker, PostgreSQL, Linux, Maven, JUnit, SonarQube, gcov, Vercel, UML (Papyrus)

EXPERIENCE

Tuffee

May 2025 – Aug. 2025

Customer Experience Associate

- Handled front-line service at an **industrial supplier of hydraulic and pneumatic parts**, resolving customer inquiries and order issues under time pressure
- Ran **point-of-sale transactions, inventory checks**, and **opening/closing procedures** as part of a small team

PROJECTS

TriageFlow (Mac-A-Thon) | *TypeScript, Next.js, PostgreSQL, Docker, Vercel*

Feb. 2026

- Built backend **Next.js API routes** and managed the **PostgreSQL** database (triage sessions, chat transcripts) for an **AI-assisted hospital intake app**, running Postgres in **Docker**
- Deployed to **Vercel**, configuring API routes for **dynamic rendering** so the staff dashboard always reads live database data
- Staff see a **live queue** ranked by **urgency and arrival time**, while an **LLM (Llama 3.3 via Groq)** turns patient intake chats into **structured clinical summaries** for staff to confirm

BridgeAid – 4th Place, McMaster Engineering Competition | *React, Flask, Leaflet.js, JWT*

Nov. 2025

- Built the **React** frontend and **live incident map** (**Leaflet.js**, **OpenStreetMap**) for a community disaster-response platform
- Incidents appear only after **3 unique users** report within **200 m** (haversine distance), cutting false reports; backend secured with **JWT auth**

Catan Game Simulator | *Java, Maven, JUnit, SonarQube, GitHub Actions*

Jan. 2026 – Mar. 2026

- **Top contributor (90 of 128 commits)** on a 4-person team building a turn-based Catan engine with **human and rule-based AI players**, configured from a text file
- Implemented **undo/redo** with the **Command pattern** (two history stacks), AI turn logic with **Template Method**, and move scoring with the **Visitor pattern**
- Wrote a **regex command parser** and set up the **JUnit** suite (**40 tests**, incl. boundary cases) with **SonarQube** analysis in **GitHub Actions** CI

FPGA Password Manager & ALU | *Verilog, Quartus Prime, DE1-SoC*

Nov. 2025

- Designed a **Moore finite state machine** and a **4-bit register** in **Verilog** on a **DE1-SoC FPGA** to verify a password input sequence in hardware
- Built a **3-bit ripple carry adder** and verified **circuit timing** in **Quartus Prime** simulation before deploying to the board

Anattack Study Suite | *C, Make, Bash, GitHub Actions*

Nov. 2025 – Dec. 2025

- **Top contributor (29 of 74 commits)** on a 4-person team; wrote the **login/register** and **leaderboard** modules in **C**
- Stored scores in **dynamically allocated arrays** grown with **realloc**, persisted them to file, and bounded all **scanf** input to prevent overflow